

## 9.5 Homogeneous Linear Systems with Constant Coefficients

In this section we discuss a procedure for obtaining a general solution for the homogeneous system

$$(1) \quad \mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t),$$

where  $\mathbf{A}$  is a (real) constant  $n \times n$  matrix. The general solution we seek will be defined for all  $t$  because the elements of  $\mathbf{A}$  are just constant functions, which are continuous on  $(-\infty, \infty)$  (recall Theorem 2, page 516). In Section 9.4 we showed that a general solution to (1) can be constructed from a fundamental solution set consisting of  $n$  linearly independent solutions to (1). Thus our goal is to find  $n$  such vector solutions.

In Chapter 4 we were successful in solving homogeneous linear equations with constant coefficients by guessing that the equation had a solution of the form  $e^{rt}$ . Because any scalar linear equation can be expressed as a system, it is reasonable to expect system (1) to have solutions of the form

$$\mathbf{x}(t) = e^{rt}\mathbf{u},$$

where  $r$  is a constant and  $\mathbf{u}$  is a constant vector, both of which must be determined. Substituting  $e^{rt}\mathbf{u}$  for  $\mathbf{x}(t)$  in (1) gives

$$re^{rt}\mathbf{u} = \mathbf{A}e^{rt}\mathbf{u} = e^{rt}\mathbf{A}\mathbf{u}.$$

Canceling the factor  $e^{rt}$  and rearranging terms, we find that

$$(2) \quad (\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{0},$$

where  $r\mathbf{I}$  denotes the diagonal matrix with  $r$ 's along its main diagonal.

The preceding calculation shows that  $\mathbf{x}(t) = e^{rt}\mathbf{u}$  is a solution to (1) if and only if  $r$  and  $\mathbf{u}$  satisfy equation (2). Since the trivial case,  $\mathbf{u} = \mathbf{0}$ , is of no help in finding linearly independent solutions to (1), we require that  $\mathbf{u} \neq \mathbf{0}$ . Such vectors are given a special name, as follows.

### Eigenvalues and Eigenvectors

**Definition 3.** Let  $\mathbf{A} = [a_{ij}]$  be an  $n \times n$  constant matrix. The **eigenvalues** of  $\mathbf{A}$  are those (real or complex) numbers  $r$  for which  $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{0}$  has at least one nontrivial (real or complex) solution  $\mathbf{u}$ . The corresponding nontrivial solutions  $\mathbf{u}$  are called the **eigenvectors** of  $\mathbf{A}$  associated with  $r$ .

As stated in Theorem 1 of Section 9.3, a linear homogeneous system of  $n$  algebraic equations in  $n$  unknowns has a nontrivial solution if and only if the determinant of its coefficients is zero. Hence, a necessary and sufficient condition for (2) to have a nontrivial solution is that

$$(3) \quad |\mathbf{A} - r\mathbf{I}| = 0.$$

Expanding the determinant of  $\mathbf{A} - r\mathbf{I}$  in terms of its cofactors, we find that it is an  $n$ th-degree polynomial in  $r$ ; that is,

$$(4) \quad |\mathbf{A} - r\mathbf{I}| = p(r).$$

Therefore, *finding the eigenvalues of a matrix  $\mathbf{A}$  is equivalent to finding the zeros of the polynomial  $p(r)$* . Equation (3) is called the **characteristic equation** of  $\mathbf{A}$ , and  $p(r)$  in (4) is the **characteristic polynomial** of  $\mathbf{A}$ . The characteristic equation plays a role for systems similar to the role played by the auxiliary equation for scalar equations.

Many commercially available software packages can be used to compute the eigenvalues and eigenvectors for a given matrix. Three such packages are MATLAB<sup>®</sup>, available from The MathWorks, Inc.; MATHEMATICA<sup>®</sup>, available from Wolfram Research; and MAPLESOFT<sup>®</sup>, available from Waterloo Maple Inc. Although you are encouraged to make use of such packages, the examples and most exercises in this text can be easily carried out without them. Those exercises for which a computer package is desirable are flagged with the icon .

**Example 1** Find the eigenvalues and eigenvectors of the matrix

$$\mathbf{A} := \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix}.$$

**Solution** The characteristic equation for  $\mathbf{A}$  is

$$|\mathbf{A} - r\mathbf{I}| = \begin{vmatrix} 2-r & -3 \\ 1 & -2-r \end{vmatrix} = (2-r)(-2-r) + 3 = r^2 - 1 = 0.$$

Hence the eigenvalues of  $\mathbf{A}$  are  $r_1 = 1$ ,  $r_2 = -1$ . To find the eigenvectors corresponding to  $r_1 = 1$ , we must solve  $(\mathbf{A} - r_1\mathbf{I})\mathbf{u} = \mathbf{0}$ . Substituting for  $\mathbf{A}$  and  $r_1$  gives

$$(5) \quad \begin{bmatrix} 1 & -3 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Notice that this matrix equation is equivalent to the single scalar equation  $u_1 - 3u_2 = 0$ . Therefore, the solutions to (5) are obtained by assigning an arbitrary value for  $u_2$  (say,  $u_2 = s$ ) and setting  $u_1 = 3u_2 = 3s$ . Consequently, the eigenvectors associated with  $r_1 = 1$  can be expressed as

$$(6) \quad \mathbf{u}_1 = s \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

For  $r_2 = -1$ , the equation  $(\mathbf{A} - r_2\mathbf{I})\mathbf{u} = \mathbf{0}$  becomes

$$\begin{bmatrix} 3 & -3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Solving, we obtain  $u_1 = s$  and  $u_2 = s$ , with  $s$  arbitrary. Therefore, the eigenvectors associated with the eigenvalue  $r_2 = -1$  are

$$(7) \quad \mathbf{u}_2 = s \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad \blacklozenge$$

We remark that in the above example the collection (6) of all eigenvectors associated with  $r_1 = 1$  forms a one-dimensional subspace when the zero vector is adjoined. The same is true for  $r_2 = -1$ . These subspaces are called **eigenspaces**.

**Example 2** Find the eigenvalues and eigenvectors of the matrix

$$\mathbf{A} := \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}.$$

**Solution** The characteristic equation for  $\mathbf{A}$  is

$$|\mathbf{A} - r\mathbf{I}| = \begin{vmatrix} 1-r & 2 & -1 \\ 1 & -r & 1 \\ 4 & -4 & 5-r \end{vmatrix} = 0,$$

which simplifies to  $(r-1)(r-2)(r-3) = 0$ . Hence, the eigenvalues of  $\mathbf{A}$  are  $r_1 = 1$ ,  $r_2 = 2$ , and  $r_3 = 3$ . To find the eigenvectors corresponding to  $r_1 = 1$ , we set  $r = 1$  in  $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{0}$ . This gives

$$(8) \quad \begin{bmatrix} 0 & 2 & -1 \\ 1 & -1 & 1 \\ 4 & -4 & 4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Using elementary row operations (Gaussian elimination), we see that (8) is equivalent to the two equations

$$\begin{aligned} u_1 - u_2 + u_3 &= 0, \\ 2u_2 - u_3 &= 0. \end{aligned}$$

Thus, we can obtain the solutions to (8) by assigning an arbitrary value to  $u_2$  (say,  $u_2 = s$ ), solving  $2u_2 - u_3 = 0$  for  $u_3$  to get  $u_3 = 2s$ , and then solving  $u_1 - u_2 + u_3 = 0$  for  $u_1$  to get  $u_1 = -s$ . Hence, the eigenvectors associated with  $r_1 = 1$  are

$$(9) \quad \mathbf{u}_1 = s \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}.$$

For  $r_2 = 2$ , we solve

$$\begin{bmatrix} -1 & 2 & -1 \\ 1 & -2 & 1 \\ 4 & -4 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

in a similar fashion to obtain the eigenvectors

$$(10) \quad \mathbf{u}_2 = s \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}.$$

Finally, for  $r_3 = 3$ , we solve

$$\begin{bmatrix} -2 & 2 & -1 \\ 1 & -3 & 1 \\ 4 & -4 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

and get the eigenvectors

$$(11) \quad \mathbf{u}_3 = s \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}. \quad \blacklozenge$$

Let's return to the problem of finding a general solution to a homogeneous system of differential equations. We have already shown that  $e^{rt}\mathbf{u}$  is a solution to (1) if  $r$  is an eigenvalue and  $\mathbf{u}$  a corresponding eigenvector. The question is: Can we obtain  $n$  linearly independent solutions to the homogeneous system by finding all the eigenvalues and eigenvectors of  $\mathbf{A}$ ? The answer is yes, if  $\mathbf{A}$  has  $n$  linearly independent eigenvectors.

### $n$ Linearly Independent Eigenvectors

**Theorem 5.** Suppose the  $n \times n$  constant matrix  $\mathbf{A}$  has  $n$  linearly independent eigenvectors  $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ . Let  $r_i$  be the eigenvalue<sup>†</sup> corresponding to  $\mathbf{u}_i$ . Then

$$(12) \quad \{e^{r_1 t}\mathbf{u}_1, e^{r_2 t}\mathbf{u}_2, \dots, e^{r_n t}\mathbf{u}_n\}$$

is a fundamental solution set (and  $\mathbf{X}(t) = [e^{r_1 t}\mathbf{u}_1 \ e^{r_2 t}\mathbf{u}_2 \ \cdots \ e^{r_n t}\mathbf{u}_n]$  is a fundamental matrix) on  $(-\infty, \infty)$  for the homogeneous system  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ . Consequently, a general solution of  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  is

$$(13) \quad \mathbf{x}(t) = c_1 e^{r_1 t}\mathbf{u}_1 + c_2 e^{r_2 t}\mathbf{u}_2 + \cdots + c_n e^{r_n t}\mathbf{u}_n,$$

where  $c_1, \dots, c_n$  are arbitrary constants.

**Proof.** As we have seen, the vector functions listed in (12) are solutions to the homogeneous system. Moreover, their Wronskian is

$$W(t) = \det[e^{r_1 t}\mathbf{u}_1, \dots, e^{r_n t}\mathbf{u}_n] = e^{(r_1 + \cdots + r_n)t} \det[\mathbf{u}_1, \dots, \mathbf{u}_n].$$

Since the eigenvectors are assumed to be linearly independent, it follows from Theorem 1 in Section 9.3 that  $\det[\mathbf{u}_1, \dots, \mathbf{u}_n]$  is not zero. Hence the Wronskian  $W(t)$  is never zero. This shows that (12) is a fundamental solution set, and consequently a general solution is given by (13). ♦

An application of Theorem 5 is given in the next example.

**Example 3** Find a general solution of

$$(14) \quad \mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t), \text{ where } \mathbf{A} = \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix}.$$

**Solution** In Example 1 we showed that the matrix  $\mathbf{A}$  has eigenvalues  $r_1 = 1$  and  $r_2 = -1$ . Taking, say,  $s = 1$  in equations (6) and (7), we get the corresponding eigenvectors

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Because  $\mathbf{u}_1$  and  $\mathbf{u}_2$  are linearly independent, it follows from Theorem 5 that a general solution to (14) is

$$(15) \quad \mathbf{x}(t) = c_1 e^t \begin{bmatrix} 3 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad \blacklozenge$$

<sup>†</sup>The eigenvalues  $r_1, \dots, r_n$  may be real or complex and need not be distinct. In this section the cases we discuss have real eigenvalues. We consider complex eigenvalues in Section 9.6.

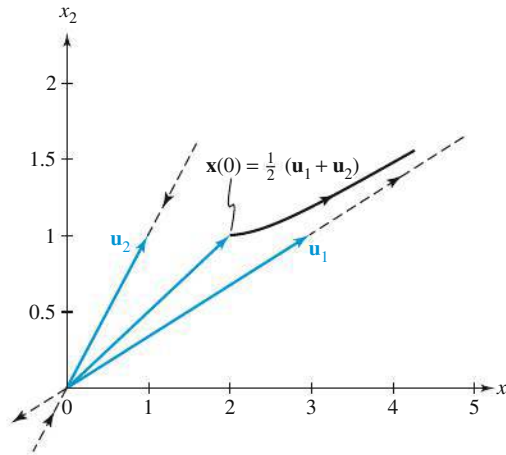


Figure 9.1 Trajectories of solutions for Example 3

If we sum the vectors on the right-hand side of equation (15) and then write out the expressions for the components of  $\mathbf{x}(t) = \text{col}(x_1(t), x_2(t))$ , we get

$$\begin{aligned}x_1(t) &= 3c_1e^t + c_2e^{-t}, \\x_2(t) &= c_1e^t + c_2e^{-t}.\end{aligned}$$

This is the familiar form of a general solution for a system, as discussed in Section 5.2.

Example 3 nicely illustrates the geometric role played by the eigenvectors  $\mathbf{u}_1$  and  $\mathbf{u}_2$ . If the initial vector  $\mathbf{x}(0)$  is a scalar multiple of  $\mathbf{u}_1$  (i.e.,  $\mathbf{x}(0) = c_1\mathbf{u}_1$ ), then the vector solution to the system,  $\mathbf{x}(t) = c_1e^t\mathbf{u}_1$ , will always have the same or opposite direction as  $\mathbf{u}_1$ . That is, it will lie along the straight line determined by  $\mathbf{u}_1$  (see Figure 9.1). Furthermore, the trajectory of this solution, as  $t$  increases, will tend to infinity, since the corresponding eigenvalue  $r_1 = 1$  is positive (observe the  $e^t$  term). A similar assertion holds if the initial vector is a scalar multiple of  $\mathbf{u}_2$ , except that since  $r_2 = -1$  is negative, the trajectory  $\mathbf{x}(t) = c_2e^{-t}\mathbf{u}_2$  will approach the origin as  $t$  increases (because of  $e^{-t}$ ). For an initial vector that involves both  $\mathbf{u}_1$  and  $\mathbf{u}_2$ , such as  $\mathbf{x}(0) = \frac{1}{2}(\mathbf{u}_1 + \mathbf{u}_2)$ , the resulting trajectory is a blend of the above motions, with the contribution due to the larger eigenvalue  $r_1 = 1$  dominating as  $t$  increases; see Figure 9.1.

The straight-line trajectories in the  $x_1x_2$ -plane (the phase plane), then, point along the directions of the eigenvectors of the matrix  $\mathbf{A}$ . (See Section 5.4, Figure 5.11, page 266, for example.)

A useful property of eigenvectors that concerns their linear independence is stated in the next theorem.

### Linear Independence of Eigenvectors

**Theorem 6.** If  $r_1, \dots, r_m$  are *distinct* eigenvalues for the matrix  $\mathbf{A}$  and  $\mathbf{u}_i$  is an eigenvector associated with  $r_i$ , then  $\mathbf{u}_1, \dots, \mathbf{u}_m$  are linearly independent.

**Proof.** Let's first treat the case  $m = 2$ . Suppose, to the contrary, that  $\mathbf{u}_1$  and  $\mathbf{u}_2$  are linearly dependent so that

$$(16) \quad \mathbf{u}_1 = c\mathbf{u}_2$$

for some constant  $c$ . Multiplying both sides of (16) by  $\mathbf{A}$  and using the fact that  $\mathbf{u}_1$  and  $\mathbf{u}_2$  are eigenvectors with corresponding eigenvalues  $r_1$  and  $r_2$ , we obtain

$$(17) \quad r_1 \mathbf{u}_1 = cr_2 \mathbf{u}_2.$$

Next we multiply (16) by  $r_2$  and then subtract from (17) to get

$$(r_1 - r_2) \mathbf{u}_1 = \mathbf{0}.$$

Since  $\mathbf{u}_1$  is not the zero vector, we must have  $r_1 = r_2$ . But this violates the assumption that the eigenvalues are distinct! Hence  $\mathbf{u}_1$  and  $\mathbf{u}_2$  are linearly independent.

The cases  $m > 2$  follow by induction. The details of the proof are left as Problem 48. ♦

Combining Theorems 5 and 6, we get the following corollary.

### $n$ Distinct Eigenvalues

**Corollary 1.** If the  $n \times n$  constant matrix  $\mathbf{A}$  has  $n$  distinct eigenvalues  $r_1, \dots, r_n$  and  $\mathbf{u}_i$  is an eigenvector associated with  $r_i$ , then

$$\{e^{r_1 t} \mathbf{u}_1, \dots, e^{r_n t} \mathbf{u}_n\}$$

is a fundamental solution set for the homogeneous system  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ .

**Example 4** Solve the initial value problem

$$(18) \quad \mathbf{x}'(t) = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}.$$

**Solution** In Example 2 we showed that the  $3 \times 3$  coefficient matrix  $\mathbf{A}$  has the three distinct eigenvalues  $r_1 = 1$ ,  $r_2 = 2$ , and  $r_3 = 3$ . If we set  $s = 1$  in equations (9), (10), and (11), we obtain the corresponding eigenvectors

$$\mathbf{u}_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix},$$

whose linear independence is guaranteed by Theorem 6. Hence, a general solution to (18) is

$$(19) \quad \begin{aligned} \mathbf{x}(t) &= c_1 e^t \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} + c_3 e^{3t} \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} -e^t & -2e^{2t} & -e^{3t} \\ e^t & e^{2t} & e^{3t} \\ 2e^t & 4e^{2t} & 4e^{3t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}. \end{aligned}$$

To satisfy the initial condition in (18), we solve

$$\mathbf{x}(0) = \begin{bmatrix} -1 & -2 & -1 \\ 1 & 1 & 1 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

and find that  $c_1 = 0$ ,  $c_2 = 1$ , and  $c_3 = -1$ . Inserting these values into (19) gives the desired solution. ♦

There is a special class of  $n \times n$  matrices that *always* have  $n$  linearly independent eigenvectors. These are the real symmetric matrices.

### Real Symmetric Matrices

**Definition 4.** A **real symmetric matrix**  $\mathbf{A}$  is a matrix with real entries that satisfies  $\mathbf{A}^T = \mathbf{A}$ .

Taking the transpose of a matrix interchanges its rows and columns. Doing this is equivalent to “flipping” the matrix about its main diagonal. Consequently,  $\mathbf{A}^T = \mathbf{A}$  if and only if  $\mathbf{A}$  is symmetric about its main diagonal.

If  $\mathbf{A}$  is an  $n \times n$  real symmetric matrix, it is known<sup>†</sup> that all its eigenvalues are real and that there always exist  $n$  linearly independent eigenvectors. In such a case, Theorem 5 applies and a general solution to  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  is given by (13).

**Example 5** Find a general solution of

$$(20) \quad \mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t), \text{ where } \mathbf{A} = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}.$$

**Solution**  $\mathbf{A}$  is symmetric, so we are assured that  $\mathbf{A}$  has three linearly independent eigenvectors. To find them, we first compute the characteristic equation for  $\mathbf{A}$ :

$$|\mathbf{A} - r\mathbf{I}| = \begin{vmatrix} 1-r & -2 & 2 \\ -2 & 1-r & 2 \\ 2 & 2 & 1-r \end{vmatrix} = -(r-3)^2(r+3) = 0.$$

Thus the eigenvalues of  $\mathbf{A}$  are  $r_1 = r_2 = 3$  and  $r_3 = -3$ .

Notice that the eigenvalue  $r = 3$  has multiplicity 2 when considered as a root of the characteristic equation. Therefore, we must find *two* linearly independent eigenvectors associated with  $r = 3$ . Substituting  $r = 3$  in  $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{0}$  gives

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -2 & 2 \\ 2 & 2 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

This system is equivalent to the single equation  $-u_1 - u_2 + u_3 = 0$ , so we can obtain its solutions by assigning an arbitrary value to  $u_2$ , say  $u_2 = v$ , and an arbitrary value to  $u_3$ , say  $u_3 = s$ . Solving for  $u_1$ , we find  $u_1 = u_3 - u_2 = s - v$ . Therefore, the eigenvectors associated with  $r_1 = r_2 = 3$  can be expressed as

$$\mathbf{u} = \begin{bmatrix} s-v \\ v \\ s \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + v \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

<sup>†</sup>See *Fundamentals of Matrix Analysis with Applications*, by Edward Barry Saff and Arthur David Snider (John Wiley & Sons, Hoboken, New Jersey, 2016).

By first taking  $s = 1, v = 0$  and then taking  $s = 0, v = 1$ , we get the two linearly independent eigenvectors

$$(21) \quad \mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

For  $r_3 = -3$ , we solve

$$(\mathbf{A} + 3\mathbf{I})\mathbf{u} = \begin{bmatrix} 4 & -2 & 2 \\ -2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

to obtain the eigenvectors  $\text{col}(-s, -s, s)$ . Taking  $s = 1$  gives

$$\mathbf{u}_3 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}.$$

Since the eigenvectors  $\mathbf{u}_1, \mathbf{u}_2$ , and  $\mathbf{u}_3$  are linearly independent, a general solution to (20) is

$$\mathbf{x}(t) = c_1 e^{3t} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_3 e^{-3t} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}. \quad \blacklozenge$$

*If a matrix  $\mathbf{A}$  is not symmetric, it is possible for  $\mathbf{A}$  to have a repeated eigenvalue but not to have two linearly independent corresponding eigenvectors. In particular, the matrix*

$$(22) \quad \mathbf{A} = \begin{bmatrix} 1 & -1 \\ 4 & -3 \end{bmatrix}$$

has the repeated eigenvalue  $r_1 = r_2 = -1$ , but Problem 35 shows that all the eigenvectors associated with  $r = -1$  are of the form  $\mathbf{u} = s \text{col}(1, 2)$ . Consequently, no two eigenvectors are linearly independent.

A procedure for finding a general solution in such a case is illustrated in Problems 35–40, but the underlying theory is deferred to Section 9.8, where we discuss the matrix exponential.

*A final note.* If an  $n \times n$  matrix  $\mathbf{A}$  has  $n$  linearly independent eigenvectors  $\mathbf{u}_i$  with eigenvalues  $r_i$ , a little inspection reveals that property (2) is expressed columnwise by the equation

$$(23) \quad \begin{bmatrix} \mathbf{A} \end{bmatrix} \begin{bmatrix} \vdots & \vdots & \cdots & \vdots \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \\ \vdots & \vdots & \cdots & \vdots \end{bmatrix} = \begin{bmatrix} \vdots & \vdots & \cdots & \vdots \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \\ \vdots & \vdots & \cdots & \vdots \end{bmatrix} \begin{bmatrix} r_1 & 0 & \cdots & 0 \\ 0 & r_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_n \end{bmatrix}$$

or  $\mathbf{AU} = \mathbf{UD}$ , where  $\mathbf{U}$  is the matrix whose column vectors are eigenvectors and  $\mathbf{D}$  is a diagonal matrix whose diagonal entries are the eigenvalues. Since  $\mathbf{U}$ 's columns are independent,  $\mathbf{U}$  is invertible and we can write

$$(24) \quad \mathbf{A} = \mathbf{UDU}^{-1} \text{ or } \mathbf{D} = \mathbf{U}^{-1}\mathbf{AU},$$

and we say that  $\mathbf{A}$  is *diagonalizable*. [In this context equation (24) expresses a *similarity transformation*.] Because the argument that leads from (2) to (23) to (24) can be reversed, we have a new characterization: *An  $n \times n$  matrix has  $n$  linearly independent eigenvectors if, and only if, it is diagonalizable.*



## 9.5 EXERCISES

In Problems 1–8, find the eigenvalues and eigenvectors of the given matrix.

1.  $\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$

2.  $\begin{bmatrix} 6 & -3 \\ 2 & 1 \end{bmatrix}$

3.  $\begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$

4.  $\begin{bmatrix} 1 & 5 \\ 1 & -3 \end{bmatrix}$

5.  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{bmatrix}$

6.  $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

7.  $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 1 \\ 0 & 2 & 4 \end{bmatrix}$

8.  $\begin{bmatrix} -3 & 1 & 0 \\ 0 & -3 & 1 \\ 4 & -8 & 2 \end{bmatrix}$

In Problems 9 and 10, some of the eigenvalues of the given matrix are complex. Find all the eigenvalues and eigenvectors.

9.  $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

10.  $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

In Problems 11–16, find a general solution of the system  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$  for the given matrix  $\mathbf{A}$ .

11.  $\mathbf{A} = \begin{bmatrix} -1 & \frac{3}{4} \\ -5 & 3 \end{bmatrix}$

12.  $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 12 & 1 \end{bmatrix}$

13.  $\mathbf{A} = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 0 & 3 \\ 2 & 3 & 0 \end{bmatrix}$

14.  $\mathbf{A} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & -1 \end{bmatrix}$

15.  $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 2 & 1 & 2 \end{bmatrix}$

16.  $\mathbf{A} = \begin{bmatrix} -7 & 0 & 6 \\ 0 & 5 & 0 \\ 6 & 0 & 2 \end{bmatrix}$

17. Consider the system  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ ,  $t \geq 0$ , with

$$\mathbf{A} = \begin{bmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}.$$

- Show that the matrix  $\mathbf{A}$  has eigenvalues  $r_1 = 2$  and  $r_2 = -2$  with corresponding eigenvectors  $\mathbf{u}_1 = \text{col}(\sqrt{3}, 1)$  and  $\mathbf{u}_2 = \text{col}(1, -\sqrt{3})$ .
- Sketch the trajectory of the solution having initial vector  $\mathbf{x}(0) = -\mathbf{u}_1$ .
- Sketch the trajectory of the solution having initial vector  $\mathbf{x}(0) = \mathbf{u}_2$ .
- Sketch the trajectory of the solution having initial vector  $\mathbf{x}(0) = \mathbf{u}_2 - \mathbf{u}_1$ .

18. Consider the system  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ ,  $t \geq 0$ , with

$$\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}.$$

- Show that the matrix  $\mathbf{A}$  has eigenvalues  $r_1 = -1$  and  $r_2 = -3$  with corresponding eigenvectors  $\mathbf{u}_1 = \text{col}(1, 1)$  and  $\mathbf{u}_2 = \text{col}(1, -1)$ .
- Sketch the trajectory of the solution having initial vector  $\mathbf{x}(0) = \mathbf{u}_1$ .
- Sketch the trajectory of the solution having initial vector  $\mathbf{x}(0) = -\mathbf{u}_2$ .
- Sketch the trajectory of the solution having initial vector  $\mathbf{x}(0) = \mathbf{u}_1 - \mathbf{u}_2$ .

In Problems 19–24, find a fundamental matrix for the system  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$  for the given matrix  $\mathbf{A}$ .

19.  $\mathbf{A} = \begin{bmatrix} -1 & 1 \\ 8 & 1 \end{bmatrix}$

20.  $\mathbf{A} = \begin{bmatrix} 5 & 4 \\ -1 & 0 \end{bmatrix}$

21.  $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & -14 & 7 \end{bmatrix}$

22.  $\mathbf{A} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{bmatrix}$

23.  $\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 & -1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 7 \end{bmatrix}$

24.  $\mathbf{A} = \begin{bmatrix} 4 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$

25. Using matrix algebra techniques, find a general solution of the system

$$x' = x + 2y - z,$$

$$y' = x + z,$$

$$z' = 4x - 4y + 5z.$$

26. Using matrix algebra techniques, find a general solution of the system

$$x' = 3x - 4y,$$

$$y' = 4x - 7y.$$



In Problems 27–30, use a linear algebra software package such as MATLAB®, MAPLESOFT®, or MATHEMATICA® to compute the required eigenvalues and eigenvectors and then give a fundamental matrix for the system  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$  for the given matrix  $\mathbf{A}$ .

27.  $\mathbf{A} = \begin{bmatrix} 0 & 1.1 & 0 \\ 0 & 0 & 1.3 \\ 0.9 & 1.1 & -6.9 \end{bmatrix}$

28.  $\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 1 & 0 \\ 3 & 3 & 3 \end{bmatrix}$

29.  $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & -6 & 3 & 3 \end{bmatrix}$

$$30. \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -2 & 4 \end{bmatrix}$$

In Problems 31–34, solve the given initial value problem.

$$31. \mathbf{x}'(t) = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$32. \mathbf{x}'(t) = \begin{bmatrix} 6 & -3 \\ 2 & 1 \end{bmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{bmatrix} -10 \\ -6 \end{bmatrix}$$

$$33. \mathbf{x}'(t) = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$$

$$34. \mathbf{x}'(t) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}$$

35. (a) Show that the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 4 & -3 \end{bmatrix}$$

has the repeated eigenvalue  $r = -1$  and that all the eigenvectors are of the form  $\mathbf{u} = s \operatorname{col}(1, 2)$ .

- (b) Use the result of part (a) to obtain a nontrivial solution  $\mathbf{x}_1(t)$  to the system  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ .  
 (c) To obtain a second linearly independent solution to  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ , try  $\mathbf{x}_2(t) = te^{-t}\mathbf{u}_1 + e^{-t}\mathbf{u}_2$ . [Hint: Substitute  $\mathbf{x}_2$  into the system  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  and derive the relations

$$(\mathbf{A} + \mathbf{I})\mathbf{u}_1 = \mathbf{0}, \quad (\mathbf{A} + \mathbf{I})\mathbf{u}_2 = \mathbf{u}_1.$$

Since  $\mathbf{u}_1$  must be an eigenvector, set  $\mathbf{u}_1 = \operatorname{col}(1, 2)$  and solve for  $\mathbf{u}_2$ .]

- (d) What is  $(\mathbf{A} + \mathbf{I})^2\mathbf{u}_2$ ? (In Section 9.8,  $\mathbf{u}_2$  will be identified as a *generalized eigenvector*.)

36. Use the method discussed in Problem 35 to find a general solution to the system

$$\mathbf{x}'(t) = \begin{bmatrix} 5 & -3 \\ 3 & -1 \end{bmatrix} \mathbf{x}(t).$$

37. (a) Show that the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 6 \\ 0 & 2 & 5 \\ 0 & 0 & 2 \end{bmatrix}$$

has the repeated eigenvalue  $r = 2$  with multiplicity 3 and that all the eigenvectors of  $\mathbf{A}$  are of the form  $\mathbf{u} = s \operatorname{col}(1, 0, 0)$ .

- (b) Use the result of part (a) to obtain a solution to the system  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  of the form  $\mathbf{x}_1(t) = e^{2t}\mathbf{u}_1$ .

- (c) To obtain a second linearly independent solution to  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ , try  $\mathbf{x}_2(t) = te^{2t}\mathbf{u}_1 + e^{2t}\mathbf{u}_2$ . [Hint: Show that  $\mathbf{u}_1$  and  $\mathbf{u}_2$  must satisfy

$$(\mathbf{A} - 2\mathbf{I})\mathbf{u}_1 = \mathbf{0}, \quad (\mathbf{A} - 2\mathbf{I})\mathbf{u}_2 = \mathbf{u}_1.]$$

- (d) To obtain a third linearly independent solution to  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ , try

$$\mathbf{x}_3(t) = \frac{t^2}{2}e^{2t}\mathbf{u}_1 + te^{2t}\mathbf{u}_2 + e^{2t}\mathbf{u}_3.$$

[Hint: Show that  $\mathbf{u}_1$ ,  $\mathbf{u}_2$ , and  $\mathbf{u}_3$  must satisfy

$$(\mathbf{A} - 2\mathbf{I})\mathbf{u}_1 = \mathbf{0}, \quad (\mathbf{A} - 2\mathbf{I})\mathbf{u}_2 = \mathbf{u}_1,$$

$$(\mathbf{A} - 2\mathbf{I})\mathbf{u}_3 = \mathbf{u}_2.]$$

- (e) Show that  $(\mathbf{A} - 2\mathbf{I})^2\mathbf{u}_2 = (\mathbf{A} - 2\mathbf{I})^3\mathbf{u}_3 = \mathbf{0}$ .

38. Use the method discussed in Problem 37 to find a general solution to the system

$$\mathbf{x}'(t) = \begin{bmatrix} 3 & -2 & 1 \\ 2 & -1 & 1 \\ -4 & 4 & 1 \end{bmatrix} \mathbf{x}(t).$$

39. (a) Show that the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ -2 & -2 & -1 \end{bmatrix}$$

has the repeated eigenvalue  $r = 1$  of multiplicity 3 and that all the eigenvectors of  $\mathbf{A}$  are of the form  $\mathbf{u} = s \operatorname{col}(-1, 1, 0) + v \operatorname{col}(-1, 0, 1)$ .

- (b) Use the result of part (a) to obtain two linearly independent solutions to the system  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  of the form

$$\mathbf{x}_1(t) = e^t\mathbf{u}_1 \quad \text{and} \quad \mathbf{x}_2(t) = e^t\mathbf{u}_2.$$

- (c) To obtain a third linearly independent solution to  $\mathbf{x}' = \mathbf{A}\mathbf{x}$ , try  $\mathbf{x}_3(t) = te^t\mathbf{u}_3 + e^t\mathbf{u}_4$ . [Hint: Show that  $\mathbf{u}_3$  and  $\mathbf{u}_4$  must satisfy

$$(\mathbf{A} - \mathbf{I})\mathbf{u}_3 = \mathbf{0}, \quad (\mathbf{A} - \mathbf{I})\mathbf{u}_4 = \mathbf{u}_3.$$

Choose  $\mathbf{u}_3$ , an eigenvector of  $\mathbf{A}$ , so that you can solve for  $\mathbf{u}_4$ .]

- (d) What is  $(\mathbf{A} - \mathbf{I})^2\mathbf{u}_4$ ?

40. Use the method discussed in Problem 39 to find a general solution to the system

$$\mathbf{x}'(t) = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 7 & -4 \\ 0 & 9 & -5 \end{bmatrix} \mathbf{x}(t).$$

41. Use the substitution  $x_1 = y$ ,  $x_2 = y'$  to convert the linear equation  $ay'' + by' + cy = 0$ , where  $a$ ,  $b$ , and  $c$  are constants, into a normal system. Show that the characteristic equation for this system is the same as the auxiliary equation for the original equation.

42. (a) Show that the Cauchy–Euler equation  $at^2y'' + bty' + cy = 0$  can be written as a **Cauchy–Euler system**

$$(25) \quad t\mathbf{x}' = \mathbf{A}\mathbf{x}$$

with a constant coefficient matrix  $\mathbf{A}$ , by setting  $x_1 = y/t$  and  $x_2 = y'$ .

- (b) Show that for  $t > 0$  any system of the form (25) with  $\mathbf{A}$  an  $n \times n$  constant matrix has nontrivial solutions of the form  $\mathbf{x}(t) = t^r \mathbf{u}$  if and only if  $r$  is an eigenvalue of  $\mathbf{A}$  and  $\mathbf{u}$  is a corresponding eigenvector.

In Problems 43 and 44, use the result of Problem 42 to find a general solution of the given system.

43.  $t\mathbf{x}'(t) = \begin{bmatrix} 1 & 3 \\ -1 & 5 \end{bmatrix} \mathbf{x}(t), \quad t > 0$

44.  $t\mathbf{x}'(t) = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \mathbf{x}(t), \quad t > 0$

45. **Mixing Between Interconnected Tanks.** Two tanks, each holding 50 L of liquid, are interconnected by pipes with liquid flowing from tank A into tank B at a rate of 4 L/min and from tank B into tank A at 1 L/min (see Figure 9.2). The liquid inside each tank is kept well stirred. Pure water flows into tank A at a rate of 3 L/min, and the solution flows out of tank B at 3 L/min. If, initially, tank A contains 2.5 kg of salt and tank B contains no salt (only water), determine the mass of salt in each tank at time  $t \geq 0$ . Graph on the same axes the two quantities  $x_1(t)$  and  $x_2(t)$ , where  $x_1(t)$  is the mass of salt in tank A and  $x_2(t)$  is the mass in tank B.

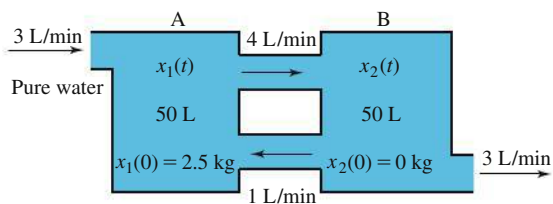


Figure 9.2 Mixing problem for interconnected tanks

46. **Mixing with a Common Drain.** Two tanks, each holding 1 L of liquid, are connected by a pipe through which liquid flows from tank A into tank B at a rate of  $3 - \alpha$  L/min ( $0 < \alpha < 3$ ). The liquid inside each tank is kept well stirred. Pure water flows into tank A at a rate of 3 L/min. Solution flows out of tank A at  $\alpha$  L/min and out of tank B at  $3 - \alpha$  L/min. If, initially, tank B contains no salt (only water) and tank A contains 0.1 kg of salt, determine the mass of salt in each tank at time  $t \geq 0$ . How does the mass of salt in tank A depend on the choice of  $\alpha$ ? What is the maximum mass of salt in tank B? (See Figure 9.3.)

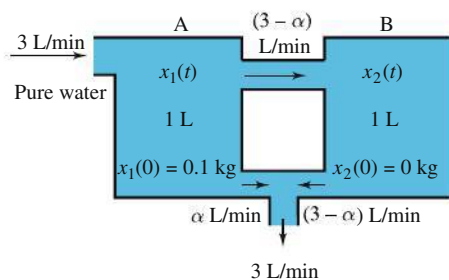


Figure 9.3 Mixing problem for a common drain,  $0 < \alpha < 3$

47. To find a general solution to the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x} = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix} \mathbf{x},$$

proceed as follows:

- (a) Use a numerical root-finding procedure to approximate the eigenvalues.  
 (b) If  $r$  is an eigenvalue, then let  $\mathbf{u} = \text{col}(u_1, u_2, u_3)$  be an eigenvector associated with  $r$ . To solve for  $\mathbf{u}$ , assume  $u_1 = 1$ . (If not  $u_1$ , then either  $u_2$  or  $u_3$  may be chosen to be 1. Why?) Now solve the system

$$(\mathbf{A} - r\mathbf{I}) \begin{bmatrix} 1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- for  $u_2$  and  $u_3$ . Use this procedure to find approximations for three linearly independent eigenvectors for  $\mathbf{A}$ .  
 (c) Use these approximations to give a general solution to the system.
48. To complete the proof of Theorem 6, page 527, assume the induction hypothesis that  $\mathbf{u}_1, \dots, \mathbf{u}_k$ ,  $2 \leq k$ , are linearly independent.

- (a) Show that if

$$c_1\mathbf{u}_1 + \dots + c_k\mathbf{u}_k + c_{k+1}\mathbf{u}_{k+1} = \mathbf{0},$$

then

$$c_1(r_1 - r_{k+1})\mathbf{u}_1 + \dots + c_k(r_k - r_{k+1})\mathbf{u}_k = \mathbf{0}.$$

- (b) Use the result of part (a) and the induction hypothesis to conclude that  $\mathbf{u}_1, \dots, \mathbf{u}_{k+1}$  are linearly independent. The theorem follows by induction.

49. **Stability.** A homogeneous system  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  with constant coefficients is **stable** if it has a fundamental matrix whose entries all remain bounded as  $t \rightarrow +\infty$ . (It will follow from Theorem 9 in Section 9.8 that if one fundamental matrix of the system has this property, then all fundamental matrices for the system do.) Otherwise, the system is **unstable**. A stable system is **asymptotically stable** if all solutions approach the zero solution as  $t \rightarrow +\infty$ . Stability is discussed in more detail in Chapter 12.<sup>†</sup>

<sup>†</sup>All references to Chapters 11–13 refer to the expanded text, *Fundamentals of Differential Equations and Boundary Value Problems*, 7th ed.

- (a) Show that if  $\mathbf{A}$  has all distinct real eigenvalues, then  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$  is stable if and only if all eigenvalues are nonpositive.
- (b) Show that if  $\mathbf{A}$  has all distinct real eigenvalues, then  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$  is asymptotically stable if and only if all eigenvalues are negative.
- (c) Argue that in parts (a) and (b), we can replace “has distinct real eigenvalues” by “is symmetric” and the statements are still true.
50. In an ice tray, the water level in any particular ice cube cell will change at a rate proportional to the *difference* between that cell’s water level and the level in the adjacent cells.

- (a) Argue that a reasonable differentiable equation model for the water levels  $x$ ,  $y$ , and  $z$  in the simplified three-cell tray depicted in Figure 9.4 is given by
- $$x' = y - x, \quad y' = x + z - 2y, \quad z' = y - z.$$
- (b) Use eigenvectors to solve this system for the initial conditions  $x(0) = 3$ ,  $y(0) = z(0) = 0$ .

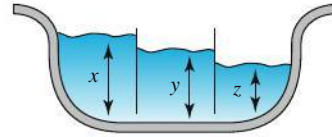


Figure 9.4 Ice tray

## 9.6 Complex Eigenvalues

In the previous section, we showed that the homogeneous system

$$(1) \quad \mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t),$$

where  $\mathbf{A}$  is a constant  $n \times n$  matrix, has a solution of the form  $\mathbf{x}(t) = e^{rt}\mathbf{u}$  if and only if  $r$  is an eigenvalue of  $\mathbf{A}$  and  $\mathbf{u}$  is a corresponding eigenvector. In this section we show how to obtain two real vector solutions to system (1) when  $\mathbf{A}$  is real and has a pair<sup>†</sup> of complex conjugate eigenvalues  $\alpha + i\beta$  and  $\alpha - i\beta$ .

Suppose  $r_1 = \alpha + i\beta$  ( $\alpha$  and  $\beta$  real numbers) is an eigenvalue of  $\mathbf{A}$  with corresponding eigenvector  $\mathbf{z} = \mathbf{a} + i\mathbf{b}$ , where  $\mathbf{a}$  and  $\mathbf{b}$  are real constant vectors. We first observe that the complex conjugate of  $\mathbf{z}$ , namely  $\bar{\mathbf{z}} := \mathbf{a} - i\mathbf{b}$ , is an eigenvector associated with the eigenvalue  $r_2 = \alpha - i\beta$ . To see this, note that taking the complex conjugate of  $(\mathbf{A} - r_1\mathbf{I})\mathbf{z} = \mathbf{0}$  yields  $(\mathbf{A} - \bar{r}_1\mathbf{I})\bar{\mathbf{z}} = \mathbf{0}$  because the conjugate of the product is the product of the conjugates and  $\mathbf{A}$  and  $\mathbf{I}$  have real entries ( $\bar{\mathbf{A}} = \mathbf{A}$ ,  $\bar{\mathbf{I}} = \mathbf{I}$ ). Since  $r_2 = \bar{r}_1$ , we see that  $\bar{\mathbf{z}}$  is an eigenvector associated with  $r_2$ . Therefore, two linearly independent complex vector solutions to (1) are

$$(2) \quad \mathbf{w}_1(t) = e^{r_1 t} \mathbf{z} = e^{(\alpha + i\beta)t} (\mathbf{a} + i\mathbf{b}),$$

$$(3) \quad \mathbf{w}_2(t) = e^{r_2 t} \bar{\mathbf{z}} = e^{(\alpha - i\beta)t} (\mathbf{a} - i\mathbf{b}).$$

As in Section 4.3, where we handled complex roots to the auxiliary equation, let's use one of these complex solutions and Euler's formula to obtain two real vector solutions. With the aid of Euler's formula, we rewrite  $\mathbf{w}_1(t)$  as

$$\begin{aligned} \mathbf{w}_1(t) &= e^{\alpha t} (\cos \beta t + i \sin \beta t) (\mathbf{a} + i\mathbf{b}) \\ &= e^{\alpha t} \{ (\cos \beta t \mathbf{a} - \sin \beta t \mathbf{b}) + i (\sin \beta t \mathbf{a} + \cos \beta t \mathbf{b}) \}. \end{aligned}$$

We have thereby expressed  $\mathbf{w}_1(t)$  in the form  $\mathbf{w}_1(t) = \mathbf{x}_1(t) + i\mathbf{x}_2(t)$ , where  $\mathbf{x}_1(t)$  and  $\mathbf{x}_2(t)$  are the two *real* vector functions

$$(4) \quad \mathbf{x}_1(t) := e^{\alpha t} \cos \beta t \mathbf{a} - e^{\alpha t} \sin \beta t \mathbf{b},$$

$$(5) \quad \mathbf{x}_2(t) := e^{\alpha t} \sin \beta t \mathbf{a} + e^{\alpha t} \cos \beta t \mathbf{b}.$$

<sup>†</sup>Recall that the complex roots of a polynomial equation with real coefficients must occur in complex conjugate pairs.